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# Diffusion-Based Makeup Transfer with Facial Region-Aware Makeup Features

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## Abstract

*Current diffusion-based makeup transfer methods commonly use the makeup information encoded by off-the-shelf foundation models (*e.g.*, CLIP) as condition to preserve the makeup style of reference image in the generation. Although effective, these works mainly have two limitations: **(1)** foundation models pre-trained for generic tasks struggle to capture makeup styles; **(2)** the makeup features of reference image are injected to the diffusion denoising model as a whole for global makeup transfer, overlooking the facial region-aware makeup features (*i.e.*, eyes, mouth, etc) and limiting the regional controllability for region-specific makeup transfer. To address these, in this work, we propose **F**acial **R**egion-**A**ware **M**akeup features (**FRAM**), which has two stages: **(1)** makeup CLIP fine-tuning; **(2)** identity and facial region-aware makeup injection. For makeup CLIP fine-tuning, unlike prior works using off-the-shelf CLIP, we synthesize annotated makeup style data using GPT-o3 and text-driven image editing model, and then use the data to train a makeup CLIP encoder through self-supervised and image-text contrastive learning. For identity and facial region-aware makeup injection, we construct before-and-after makeup image pairs from the edited images in stage 1 and then use them to learn to inject identity of source image and makeup of reference image to the diffusion denoising model for makeup transfer. Specifically, we use learnable tokens to query the makeup CLIP encoder to extract facial region-aware makeup features for makeup injection, which is learned via an attention loss to enable regional control. As for identity injection, we use a ControlNet Union to encode source image and its 3D mesh simultaneously. The experimental results verify the superiority of our regional controllability and our makeup transfer performance.*

## 1. Introduction

Makeup transfer copies the makeup style from reference face image to source image, while preserving source face's identity [46]. With the success of large-scale text-to-image (T2I) diffusion [26], recent works condition T2I diffusion models on source identity image and reference makeup image to achieve high-fidelity makeup transfer [27, 30, 33, 46].

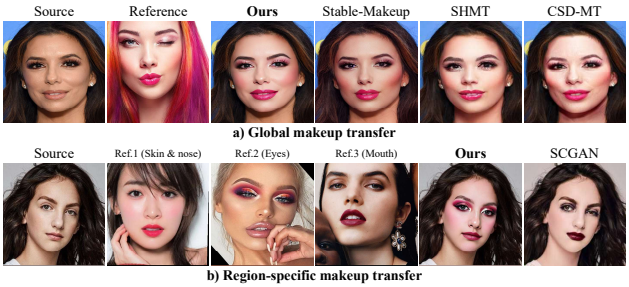



Figure 1. **Applications of our method FRAM**. Global makeup transfer copies the makeup style of reference image to source image. Region-specific makeup transfer combines the regional makeup styles (skin, eyes and mouth) of three reference images.

Current methods commonly inject the makeup features encoded by off-the-shelf foundation models (*e.g.*, CLIP [25]) to the diffusion denoising model for reference makeup preservation [33, 46]. Despite their effectiveness, these methods are mainly limited in two aspects: **(1)** foundation models pre-trained on natural images are optimized for generic tasks, and therefore struggle to capture facial makeup styles; **(2)** these works inject the makeup features of reference image to the diffusion denoising model as a whole for global makeup transfer, overlooking the facial region-aware makeup features (*i.e.*, eyes, mouth, etc) and limiting regional controllability for region-specific makeup transfer. To address these, first, unlike prior works that directly use off-the-shelf CLIP [25], we **train a makeup CLIP encoder** to encode makeup styles by synthesizing annotated makeup style data for CLIP fine-tuning. In Tab. 3, we show that our makeup CLIP encoder is a better makeup style encoder. Moreover, inspired by self-supervised facial representation learning [8], we use learnable tokens as queries for the makeup CLIP encoder to extract **facial region-aware makeup features** for makeup injection. This is learned with an attention loss, which encourages the model to look at facial regions (see cross-attention maps in Fig. 5) and enables region-specific makeup transfer (Fig. 1b and Fig. 5). This is in contrast to prior works that **inject makeup features as a whole** for global makeup transfer [33, 46].

In this work, we propose a diffusion-based facial makeup transfer framework, **F**acial **R**egion-**A**ware **M**akeup features (**FRAM**), which has two stages: **(1) makeup CLIP fine-tuning** learns a makeup CLIP encoder; **(2) identity and facial region-aware makeup injection** learns to extract identity features of source image with the ControlNet [45] and facial region-aware makeup features of reference image from our makeup CLIP, and then inject them to the diffusion denoising model for makeup transfer. For *makeup CLIP fine-tuning*, as existing makeup datasets lack labels and text descriptions, we synthesize annotated makeup style data for CLIP fine-tuning by using GPT-o3 to generate makeup style descriptions and then prompt a state-of-the-art (SOTA) text-driven image editing model (*e.g.*, FLUX.1-Kontext-dev [17]) to add makeup to face images, obtaining images with various makeup styles. Inspired by CSD [31] that measures natural image style similarity, we fine-tune CLIP to learn content-invariant makeup features using 2 objectives: self-supervised and image-text contrastive learning. Unlike CSD [31] that only adopts image contrastive learning on natural artistic images, we further incorporate the text supervision of makeup style descriptions via image-text contrastive learning on synthetic face images to learn makeup features. For *identity and facial region-aware makeup injection*, given the scarcity of paired data, we construct before-and-after makeup image pairs from the edited images in stage 1. The paired data is used to learn identity and makeup information injection for makeup transfer. We propose to extract facial region-aware makeup features to enable region-specific makeup transfer. Specifically, inspired by facial representation learning [8], we use learnable tokens as queries for our makeup CLIP encoder to output a set of "facial region makeup embeddings", each associated with a facial region. These embeddings are treated as image prompts and injected to the denoising model through cross-attention to control the sampling process. To supervise the learning of these embeddings, we align the cross-attention maps of these image prompts and the face parsing masks (produced by a face parsing model FaRL [47]) of reference image (*i.e.*, synthesized makeup image of source, which shares the same masks as source). This enables the learned queries to guide the model to apply makeup to source's facial regions **without using masks** during inference. As for identity injection, unlike Stable-Makeup [46] that uses two ControlNets [45], we use a ControlNet Union [41] to leverage the pixel-level identity information of source image and the structure guidance from 3D mesh [38] simultaneously.

Our contributions can be summarized as follows:

- We propose a diffusion-based makeup transfer framework, **F**acial **R**egion-**A**ware **M**akeup features (**FRAM**).
- We learn a makeup CLIP encoder by fine-tuning CLIP on synthesized annotated makeup style data.
- We construct before-and-after makeup image pairs to learn makeup transfer. Moreover, we make a first attempt to enable diffusion-based region-specific makeup transfer by using learnable queries to extract facial region-aware makeup features from our makeup CLIP encoder.

## 2. Related work

### 2.1. Facial makeup transfer

Makeup transfer based on GAN (Generative Adversarial Network [9]) faces challenges in transferring complex real-world makeup styles [3, 14, 15, 18, 20, 24, 34, 40, 42]. To address this, recent works leverage pre-trained diffusion model [6, 21, 22] for makeup transfer by using source identity image and reference makeup image as conditions, which are injected to the diffusion denoising model to control the sampling process [27, 30, 33, 46]. Stable-Makeup [46] uses the ControlNet [45] and off-the-shelf CLIP [25] to extract identity and makeup features, respectively. In contrast, we synthesize annotated makeup style data and fine-tune CLIP on it to learn a makeup CLIP encoder. Moreover, unlike Stable-Makeup [46] that injects CLIP features to the diffusion denoising model as a whole for global makeup transfer, we learn to extract facial region-aware makeup features from our makeup CLIP, enabling region-specific makeup transfer. Gorgeous [30] encodes makeup styles by learning text embeddings via textual inversion [7], which are used as the text condition for denoising model. In contrast, we use our makeup CLIP encoder as the makeup encoder instead of optimizing text embeddings, which eliminates the need to learn a new token for each new concept of makeup style. SHMT [33] proposes a self-supervised method to learn the identity and makeup injection using existing makeup datasets. By contrast, we synthesize before-and-after makeup image pairs for supervised learning. MAD [27] is a training-free method that blends source and reference image to obtain the noise seed so that the makeup and identity information are encoded during sampling. However, MAD [27] is less effective than the training-based methods above. Concurrent works [39, 48] also apply image editing to generate paired makeup data. However, these works simply discard the generated images with large face region misalignment while we alleviate this by aligning the face regions via affine transformation (Fig. 3). We additionally show that the aligned data leads to better performance in the supplementary. Moreover, these works lack regional controllability of several references.

### 2.2. Paired makeup data generation

Recent works use a training-free image editing model to add makeup to face images to obtain before-and-after makeup image pairs [20, 46]. However, the quality of generated makeup image is limited by the training-free model. In contrast, we use the SOTA training-based image editing model

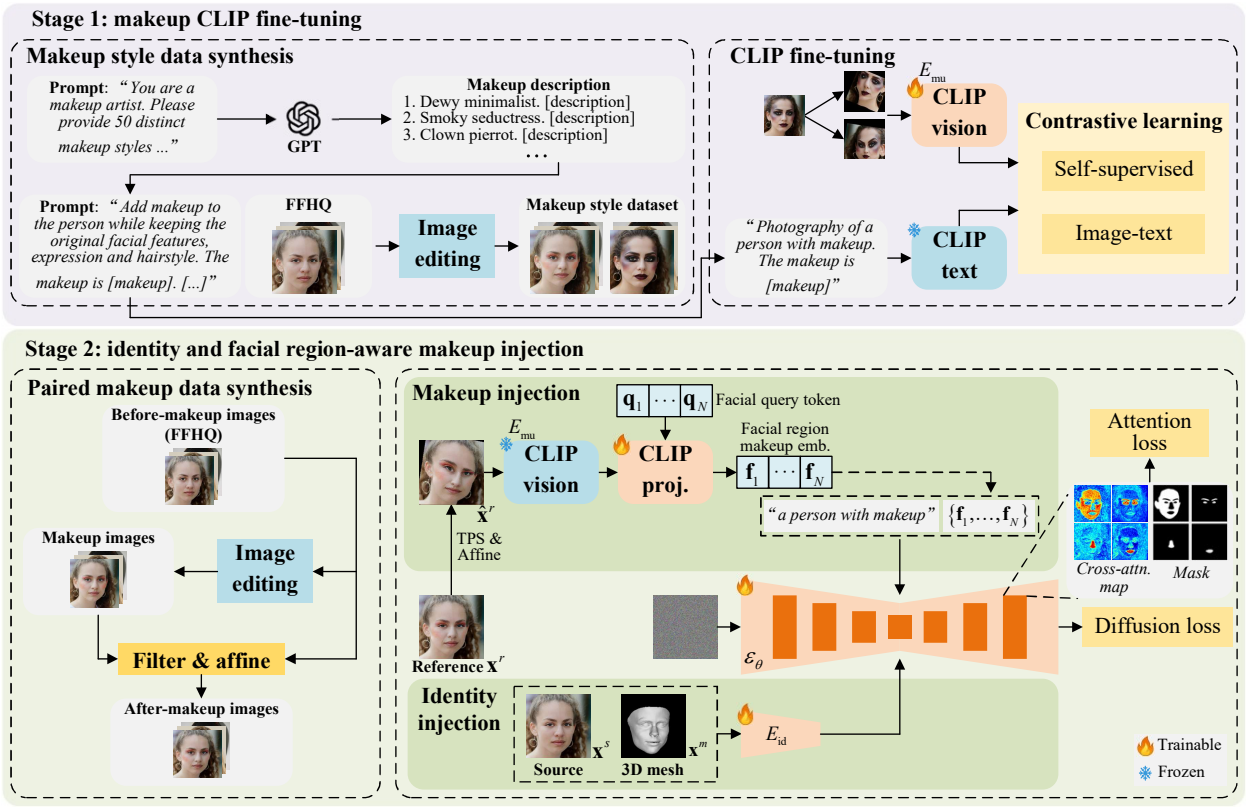



Figure 2. **Overview of our FRAM**. It has two stages: **(1)** makeup CLIP fine-tuning; **(2)** identity and facial region-aware makeup injection. For stage 1, we synthesize annotated makeup style data using GPT-o3 and a SOTA text-driven image editing model (*e.g.*, FLUX.1-Kontext-dev [17]), and then use it to learn a makeup CLIP vision encoder $E_{\text{mu}}$. For stage 2, we construct before-and-after makeup image pairs from the edited images in stage 1 and use them to learn to transfer the makeup style of reference image $\mathbf{x}^r$ to source image $\mathbf{x}^s$, *i.e.*, inject identity and makeup information to the diffusion denoising model $\epsilon_\theta$. For makeup injection, we extract facial region makeup embeddings $\{\mathbf{f}_n\}_{n=1}^N$ by querying our makeup CLIP encoder $E_{\text{mu}}$ and inject them to $\epsilon_\theta$ via cross-attention. For identity injection, we adopt a ControlNet Union [41] to encode pixel-level identity information of source image $\mathbf{x}^s$ and structure guidance from 3D mesh $\mathbf{x}^m$ simultaneously.




Figure 3. **Affine *vs*. blend face and non-face region**.

(*e.g.*, FLUX.1-Kontext-dev [17]) to edit face images. Moreover, to avoid edits on non-face region, these works commonly combine the face region of generated makeup image and non-face region of source identity image to obtain the before-and-after makeup pair, under the guidance of face parsing masks. However, the face region of generated makeup image may not be spatially aligned with the face region of source image. Thus, the composite image is distorted. To alleviate this, we align face regions between the generated makeup image and source image via affine transformation. Another work, TinyBeauty [15] uses 5 predefined makeup styles from commercial photo editor MEITU to fine-tune the diffusion model for paired data generation. FFHQ-Makeup [43] synthesizes paired data via makeup transfer on makeup styles from existing datasets. However, the diversity of makeup styles in these works is limited. In contrast, we use GPT-o3 to generate various makeup style descriptions and then prompt an image editing model to synthesize face images with diverse makeup styles. Unlike the above methods that add makeup to face, some [10, 30, 32] remove the makeup of face to obtain paired data. However, these works are suboptimal due to the limited makeup style diversity of existing makeup datasets.

### 2.3. Facial region discovery

GAN-based methods have also explored facial region discovery for makeup transfer [3, 40]. SCGAN [3] extracts makeup features for facial regions at the image level, under the guidance of face parsing masks. RamGAN [40] extracts

makeup features for facial regions by aggregating feature maps at the feature level. Despite different techniques, these methods are tailored for GAN while we propose to query our makeup CLIP encoder and inject facial region-aware makeup features to the diffusion denoising model.

## 3. Methodology

### 3.1. Overview

An overview of our FRAM is shown in Fig. 2. Given a reference makeup image $\mathbf{x}^r$ and source identity image $\mathbf{x}^s$, the goal is to leverage the diffusion denoising process to transfer the makeup style of $\mathbf{x}^r$ to $\mathbf{x}^s$, while preserving the identity of $\mathbf{x}^s$. Following common practice [26], the diffusion denoising model $\epsilon_\theta$ operates in the latent space of a Variational Autoencoder (VAE) [5]. During training, Gaussian noise $\epsilon$ is added to $\mathbf{x}^r$ to obtain noised latent $\mathbf{x}_t$ at timestep $t \in \{0, \dots, T\}$, where $\mathbf{x}_0 = \mathbf{x}^r$. $\epsilon_\theta$ is trained to predict the added noise to denoise $\mathbf{x}_t$. During sampling, $\epsilon_\theta$ recovers the clean sample $\mathbf{x}_0$ by gradually denoising the Gaussian noise via the denoising process. To control the denoising, the identity features of $\mathbf{x}^s$ and makeup features of $\mathbf{x}^r$ are injected to $\epsilon_\theta$ as conditions. The training of our FRAM has two stages: **(1)** makeup CLIP fine-tuning; **(2)** identity and facial region-aware makeup injection. For stage 1, we synthesize annotated makeup style data and use it to learn a makeup CLIP encoder $E_{\text{mu}}$ (Sec. 3.2). For stage 2, we construct before-and-after makeup image pairs and use them to learn makeup transfer by injecting identity and makeup information to $\epsilon_\theta$, *i.e.*, identity features from an identity encoder $E_{\text{id}}$ and facial region-aware makeup features from our makeup CLIP encoder $E_{\text{mu}}$ (Sec. 3.3).

### 3.2. Stage 1: makeup CLIP fine-tuning

**Makeup style data synthesis**. To synthesize annotated makeup style data with various makeup styles for CLIP fine-tuning, we prompt GPT-o3 to generate 50 distinct, diverse and detailed makeup style descriptions, covering styles from light to heavy makeup, *e.g.*, "*Dewy minimalist. [description]*", where "[description]" denotes the detailed description of makeup style "dewy minimalist". Then we use these makeup styles to prompt a SOTA text-driven image editing model (*e.g.*, FLUX.1-Kontext-dev [17]) to add makeup to face images from the dataset FFHQ [16] to obtain the makeup style images. The prompt template is "*Add makeup to the person while keeping the original facial features, expression and hairstyle. The makeup is [makeup]. Maintain the original position, background, camera angle, framing, and perspective.*", where "[makeup]" is the makeup style description from GPT-o3. The SOTA image editing model ensures the facial identity is preserved in the generation while FFHQ ensures facial diversity. More details and samples of generated makeup style descriptions and images are provided in the supplementary.

**CLIP fine-tuning**. Next, we fine-tune a CLIP vision encoder to learn a makeup encoder $E_{\text{mu}}$, using self-supervised and image-text contrastive learning. Given a face image $\mathbf{x}_i$ from the synthesized makeup style data, we augment its content to obtain two views of $\mathbf{x}_i$, *i.e.*, $\{\mathbf{x}_{i,v} \mid v \in \{1, 2\}\}$, which exhibit altered facial content while preserving the makeup style. Following [31, 46], we use Thin Plate Spline (TPS) [35], random crop, flip, and affine transformation for augmentations. Each $\mathbf{x}_{i,v}$ is fed into $E_{\text{mu}}$ to obtain the embedding, *i.e.*, $\mathbf{z}_{i,v} = E_{\text{mu}}(\mathbf{x}_{i,v})$. Let $I = \{(i, v) \mid i \in \{1, \dots, B\}, v \in \{1, 2\}\}$ be the set of indices of samples in the mini-batch. We use self-supervised contrastive learning (SSL) to fine-tune CLIP to maximize the similarity of the two views, which encourages CLIP to ignore face structure while preserving high-level semantics (*e.g.*, makeup style). The InfoNCE loss [2] is adopted for SSL:

$$\mathcal{L}_{\text{ssl}} = -\log \frac{\exp\left(f_s(\mathbf{z}_{i,1}, \mathbf{z}_{i,2})/\tau\right)}{\sum_{a \in I(i,1)} \exp\left(f_s(\mathbf{z}_{i,1}, \mathbf{z}_a)/\tau\right)}, \quad (1)$$

where $f_s(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$ denotes the cosine similarity between $\mathbf{u}$ and $\mathbf{v}$, $\tau$ is the temperature, and $I(i, 1) \equiv I \setminus \{(i, 1)\}$. The above objective only uses the image embeddings. We further align the image embedding of $E_{\text{mu}}$ with the text embedding of a pre-trained CLIP text encoder, which encodes the makeup style descriptions. The CLIP text encoder takes as input the prompt template "*Photography of a person with makeup. The makeup is [makeup]*", where "[makeup]" is the makeup style description. The image-text contrastive learning objective is:

$$\mathcal{L}_{\text{text}} = -\frac{1}{|P|} \sum_{p \in P} \log \frac{\exp\left(f_s(\mathbf{z}_{i,1}, \mathbf{z}_p^{\text{text}})/\tau\right)}{\sum_{a \in I} \exp\left(f_s(\mathbf{z}_{i,1}, \mathbf{z}_a^{\text{text}})/\tau\right)}, \quad (2)$$

where $P$ is the set of indices of positive samples (same makeup style as $\mathbf{z}_{i,1}$) in the mini-batch, $|P|$ is the number of positive samples, $\mathbf{z}_p^{\text{text}}$ and $\mathbf{z}_a^{\text{text}}$ are text embeddings for positive sample of $\mathbf{z}_{i,1}$ and sample in the batch, respectively.

The overall CLIP fine-tuning objective is as follows:

$$\mathcal{L}_{\text{clip}} = \mathcal{L}_{\text{ssl}} + \mathcal{L}_{\text{text}}, \quad (3)$$

We initialize $E_{\text{mu}}$ with pre-trained CLIP and fine-tune the last encoder layer. The learned $E_{\text{mu}}$ is used to extract the makeup features of reference image in stage 2 (Sec. 3.3).

### 3.3. Stage 2: identity and facial region-aware makeup injection

**Paired makeup data synthesis**. To construct before-and-after makeup image pairs, we conduct filtering and alignment on the pairs of FFHQ [16] and makeup style images in Sec. 3.2. First, we use GPT-5 to filter out FFHQ-makeup image pairs with non-realistic makeup images or different

facial identities and expressions. To avoid editing non-face region during image editing, prior works commonly blend the face region of generated makeup image and non-face region of source identity image to obtain the before-and-after makeup pair, under the guidance of face parsing masks [20, 46]. However, as shown in Fig. 3, image editing could cause spatial misalignment, *i.e.*, the face region of generated makeup image (2nd column) may not be spatially aligned with the face region of source image. Thus, the composite image (3rd column) is distorted. To mitigate this, we perform affine transformation based on the facial landmarks (predicted by JMLR [11]) to replace source image's face with that of generated makeup image. This aligns the face regions between the generated image and source image to obtain the after-makeup image (4th column). Next, the images with teeth and eye misalignments relative to the source image are filtered out based on the IoU (Intersection over Union) of their face parsing masks (produced by FaRL [47]). More details are in the supplementary.

With the synthesized image pairs, we learn to inject identity and makeup information to $\epsilon_\theta$ for makeup transfer.

**Makeup injection**. Inspired by face representation learning, we extract facial region-aware makeup features from $E_{\text{mu}}$. First, we follow [46] to augment the structure of $\mathbf{x}^r$ via transformations (TPS and affine) to obtain $\hat{\mathbf{x}}^r$. Then, a CLIP projector takes as input $N$ learnable tokens $\{\mathbf{q}_n\}_{n=1}^N$ (query) and the CLIP features from the last encoder layer $E_{\text{mu}}(\hat{\mathbf{x}}^r)$ (key and value) to predict $N$ "facial region makeup embeddings" $\{\mathbf{f}_n\}_{n=1}^N$, each associated with a facial region. The CLIP projector adopts the structure used in InstantID [36]. Similar to IP-Adapter [44], the embeddings $\{\mathbf{f}_n\}_{n=1}^N$ are treated as **image prompt** embeddings, allowing $\epsilon_\theta$ to inject the makeup features via cross-attention. The text prompt for $\epsilon_\theta$ is "*a person with makeup*". Let $\mathbf{A}_n^l$ be the cross-attention map of $\mathbf{f}_n$ (facial region $n$) at layer $l$, $\bar{\mathbf{A}}_n = \frac{1}{L}\sum_{l=1}^L \mathbf{A}_n^l$ be the averaged cross-attention map across $L$ layers. To supervise the learning of $\{\mathbf{f}_n\}_{n=1}^N$, we align the cross-attention maps of $\{\mathbf{f}_n\}_{n=1}^N$ and face parsing masks of $\mathbf{x}_r$ using the following attention loss $\mathcal{L}_{\text{attn}}$:

$$\mathcal{L}_{\text{attn}} = \frac{1}{NUV}\sum_{n=1}^{N}\sum_{u,v}\Big[\text{FL}(\bar{\mathbf{A}}_n[u,v], \mathbf{M}_n[u,v]) + \text{DICE}(\bar{\mathbf{A}}_n[u,v], \mathbf{M}_n[u,v])\Big], \tag{4}$$

where $U$ and $V$ are the attention map size, $[u, v]$ is the spatial location, $\mathbf{M}_n$ is the binary mask for facial region $n$ (*i.e.*, skin, eyes, nose, and mouth) predicted by a face parsing model [47]. $\mathcal{L}_{\text{attn}}$ encourages $\{\mathbf{f}_n\}_{n=1}^N$ to look at the corresponding facial regions in the cross-attention maps. $\mathcal{L}_{\text{attn}}$ adopts the binary mask loss in segmentation [1], which consists of a focal loss [19] FL() and a dice loss [23] DICE(). As shown in Sec. 3.4, the learned queries enable regional control without relying on masks during inference.

**Identity injection**. We use a ControlNet Union [41] as the identity encoder $E_{\text{id}}$ for identity injection. $E_{\text{id}}$ encodes the pixel-level information from source image $\mathbf{x}^s$ and facial structure from 3D mesh $\mathbf{x}^m$ in a single ControlNet. We reconstruct 3D mesh $\mathbf{x}^m$ from $\mathbf{x}^s$ using 3DDFA-V3 [38]. Null text is used as the prompt for $E_{\text{id}}$.

**Overall objective**. In addition to $\mathcal{L}_{\text{attn}}$ (Eq. (4)), we adopt the diffusion loss [12] to learn the joint injection of identity and makeup during the denoising process:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0,t,\epsilon}\left\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, C)\right\|_2^2, \tag{5}$$

where $C$ is the conditions, including prompt, identity and makeup features. The overall objective is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{attn}}. \tag{6}$$

Since the makeup features are injected via cross-attention, we apply LoRA (Low-Rank Adaptation) [13] to the cross-attention layers of pre-trained denoising model $\epsilon_\theta$ for fine-tuning. The LoRA layers of $\epsilon_\theta$, CLIP projector and identity encoder $E_{\text{id}}$ are jointly updated during training.

### 3.4. Regional control

Since the facial region makeup embeddings $\{\mathbf{f}_n\}_{n=1}^N$ correspond to distinct facial regions, regional control can be achieved during sampling while prior diffusion-based methods lack such controllability. In Fig. 5, we construct a new set of embeddings by selecting the region-specific embeddings from different reference images (*i.e.*, skin from reference 1, eyes from reference 2 and mouth from reference 3), which are injected to the denoising model for region-specific makeup transfer.

## 4. Experiments

### 4.1. Experimental setups

**Implementation**. The number of facial regions $N$ is set to 4, corresponding to skin, eyes, nose, and mouth. The CLIP makeup encoder $E_{\text{mu}}$ adopts ViT-L/14 [4]. Following [30], we only edit the face region via an image inpainting pipeline to avoid edits on non-face area. We use Stable Diffusion v2.1 (SDv2.1) [26] as the pre-trained diffusion model for all methods when possible and baseline's default model otherwise. For Stable-Makeup [46], we retrain it with SDv2.1. The baselines adopt the default parameters. Following [46], we evaluate on three datasets: Makeup Transfer (MT) [18], Wild-MT [14] and CPM-real [24]. We perform ablation study on CPM-real [24]. The image is resized to $512 \times 512$. More details and results can be found in the supplementary.

**Baselines**. We compare with GAN-based model (*i.e.*, CSD-MT [34]) and diffusion-based models (*i.e.*, SHMT [33], MAD [27], Gorgeous [30], and Stable-Makeup [46] (Stable-M)). Since prior diffusion-based models don't support regional control, we compare with GAN-based SCGAN [3] for region-specific makeup transfer.

Table 1. **Quantitative results**. The **red**/**blue** value indicates the top/second-ranked result, respectively.

| Method | MT [18] | | | | | Wild-MT [14] | | | | | CPM-real [24] | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ |
| **GAN-based** | | | | | | | | | | | | | | | |
| CSD-MT [34] | 0.434 | 0.585 | 0.424 | 0.146 | 4.50 | 0.428 | 0.585 | 0.430 | 0.124 | 4.41 | 0.418 | 0.522 | 0.410 | 0.128 | 4.41 |
| **Diffusion-based** | | | | | | | | | | | | | | | |
| MAD [27] | 0.328 | 0.535 | 0.805 | 0.003 | 4.26 | 0.401 | **0.541** | 0.808 | 0.002 | 4.18 | 0.317 | **0.491** | 0.788 | 0.003 | 4.18 |
| Gorgeous [30] | 0.417 | **0.652** | **0.896** | 0.003 | 4.70 | 0.472 | **0.628** | **0.887** | 0.002 | 4.64 | 0.399 | **0.624** | **0.871** | **0.002** | 4.61 |
| SHMT [33] | 0.498 | 0.372 | 0.811 | 0.012 | 4.86 | 0.516 | 0.411 | 0.820 | 0.010 | 4.78 | 0.417 | 0.408 | 0.793 | 0.013 | 4.64 |
| Stable-M [46] | **0.527** | 0.413 | 0.864 | 0.006 | **5.10** | **0.562** | 0.428 | 0.864 | **0.006** | **5.02** | **0.515** | 0.415 | 0.777 | 0.007 | **4.87** |
| FRAM (**Ours**) | **0.536** | **0.587** | **0.880** | **0.002** | **5.25** | **0.571** | 0.499 | **0.866** | **0.002** | **5.05** | **0.528** | 0.429 | **0.797** | **0.003** | **4.95** |

**Metrics**. Following [20, 30, 46], we use 5 metrics for the quantitative results: **(1)** CSD [31] measures the makeup style similarity between the generated image and reference makeup image, using the cosine similarity of a pre-trained style encoder. **(2)** ID measures the facial identity preservation between the generated image and source identity image, using the cosine similarity of a face identity encoder BlendFace [29], which is robust to makeup. **(3)** SSIM [37] measures the face structure similarity between the generation and source. **(4)** L2-M [46] measures the non-edited region difference by calculating the MSE of non-face region between the generated image and source identity image. **(5)** Aesthetic Score [28] (Aes) measures the quality of the generated image using the LAION-Aesthetics Predictor.

### 4.2. Qualitative results

In Fig. 4, we report the qualitative results on global makeup transfer. We observe: **(1)** GAN-based CSD-MT [34] faces challenges in generating high-quality images that capture makeup styles. **(2)** The training-free method MAD [27] also struggles with image quality and makeup consistency. **(3)** Gorgeous [30] fails to capture makeup styles. **(4)** Stable-Makeup [46] has inferior results in preserving identity (*e.g.*, 7$^{th}$ row) and capturing the fine-grained makeup patterns (*e.g.*, highlighted effect around the eyes in 4$^{th}$ row).

In Fig. 6, we additionally provide results on cross-domain faces (*e.g.*, cartoon, painting), demonstrating our superiority against SOTA methods.

### 4.3. Quantitative results

In Tab. 1, the quantitative results are reported. We observe: **(1)** Although MAD [27] and Gorgeous [30] exhibit strong identity preservation (high ID score), both struggle to capture the desired makeup style (low CSD score). Moreover, MAD [27] suffers from low image quality (low Aes score). This is aligned with the observation in Fig. 4. **(2)** Although Stable-Makeup [46] has strong makeup consistency results, it's inferior with identity preservation, which is also observed in Fig. 4. **(3)** Compared with prior makeup transfer methods, our FRAM has a better balance between identity preservation and makeup consistency without sacrificing the image quality.

Table 2. **User study**. The best selection ratio (%) is reported.

| CSD-MT | SHMT | Stable-Makeup | Ours |
|---|---|---|---|
| 20.5% | 5% | 16% | 58.5% |

### 4.4. User study

Following [33, 46], we conduct a user study to complement quantitative evaluation. We randomly select 20 source-reference pairs from CPM-real [24] and Wild-MT [14]. 10 participants evaluated the generated images and selected the best one based on identity preservation, makeup consistency, non-edited region difference and image quality. The results reported in Tab. 2 show that our FRAM has the best performance in terms of human perception.

Table 3. **Ablation on CLIP fine-tuning objectives**. **SSL** is self-supervised contrastive learning. **Text** is image-text contrastive learning. **Acc** is K-nearest neighbour (KNN) classification accuracy on the makeup style data, where K is 5. The first row denotes off-the-shelf CLIP [25] pre-trained for generic tasks.

| SSL | Text | Acc ↑ | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ |
|---|---|---|---|---|---|---|---|
| × | × | 61.7 | 0.461 | **0.459** | **0.799** | 0.003 | 4.88 |
| ✓ | × | 80.3 | 0.506 | 0.425 | 0.786 | 0.003 | 4.86 |
| × | ✓ | 86.9 | 0.508 | 0.406 | 0.776 | 0.003 | 4.79 |
| ✓ | ✓ | **88.2** | **0.528** | 0.429 | 0.797 | **0.003** | **4.95** |

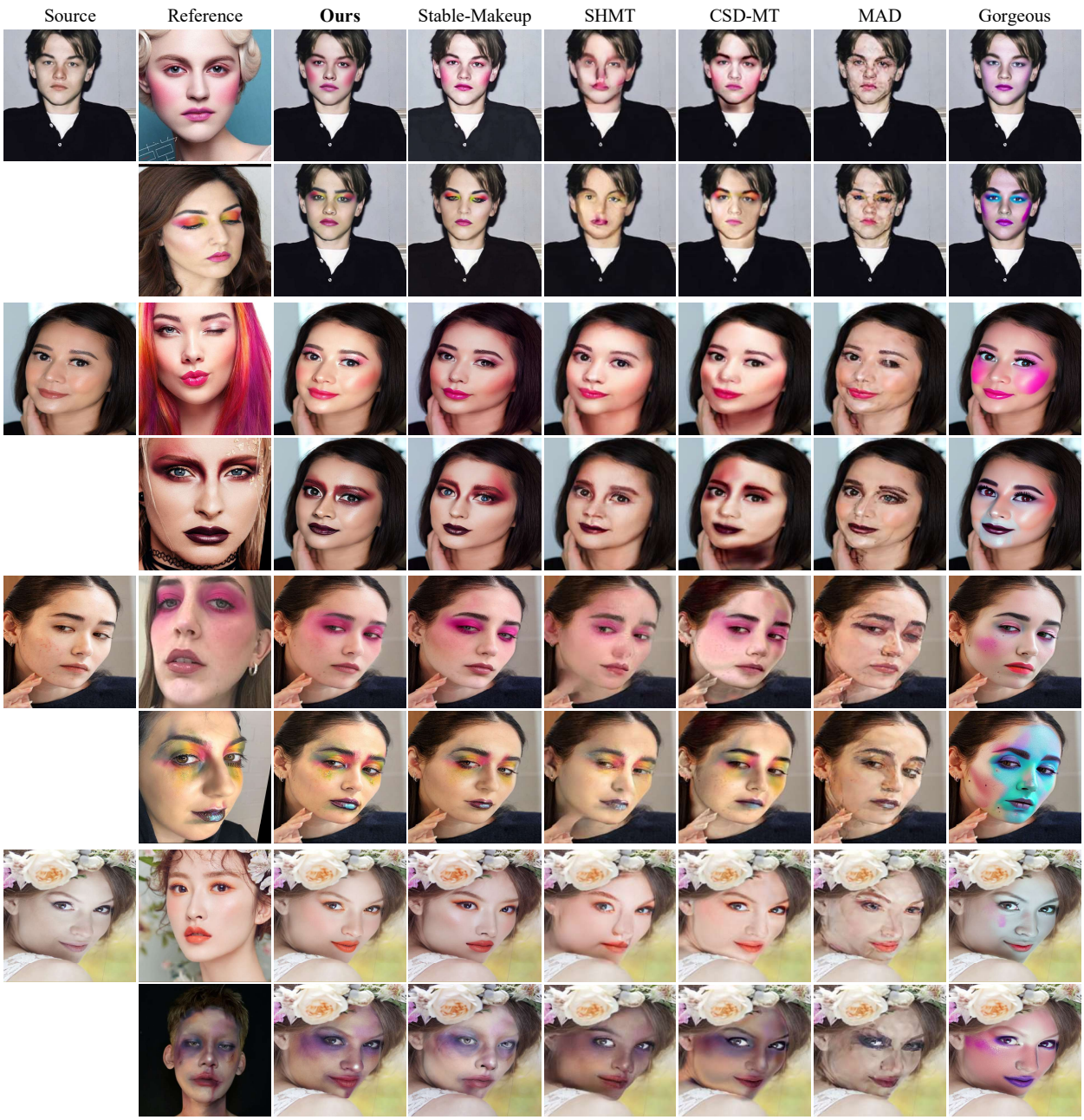



Figure 4. **Qualitative results**.

## 4.5. Ablation study

### 4.5.1. CLIP fine-tuning objectives

In Tab. 3, we ablate the CLIP fine-tuning objectives. The proposed self-supervised and image-text contrastive learning help improve makeup consistency. Altogether, they result in the best makeup consistency and better balance between identity preservation and makeup consistency. These validate our CLIP fine-tuning for makeup encoding.

### 4.5.2. Identity and makeup injection modules

In Tab. 4, we evaluate the effectiveness of identity and makeup injection modules. We observe: **(1)** LoRA fine-tuning improves makeup consistency by fusing the CLIP makeup features. **(2)** Both the pixel-level information and facial structure of 3D mesh from source image contribute to the identity preservation. These validate our identity and makeup injection modules.

We also provide qualitative ablation in Fig. 7, which is in

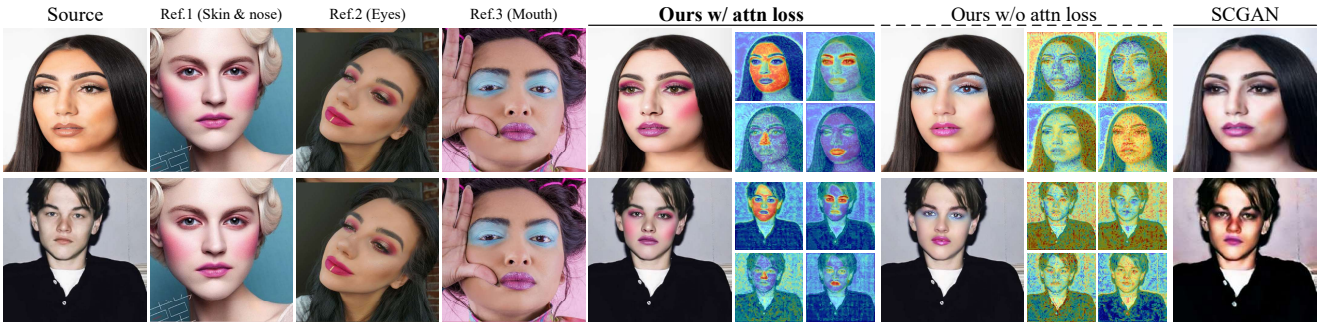



Figure 5. **Region-specific makeup transfer and qualitative ablation on attention loss**. The visualization of the cross-attention maps corresponding to the facial region makeup embeddings is next to the generated image.

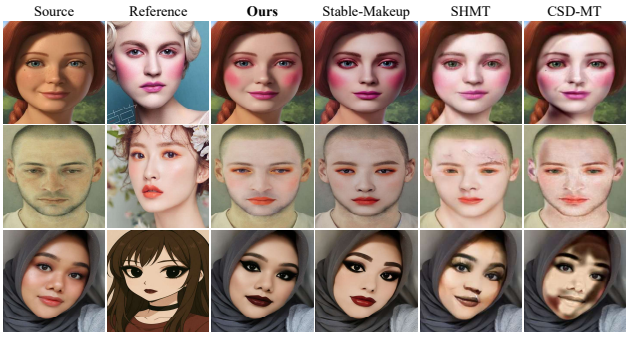
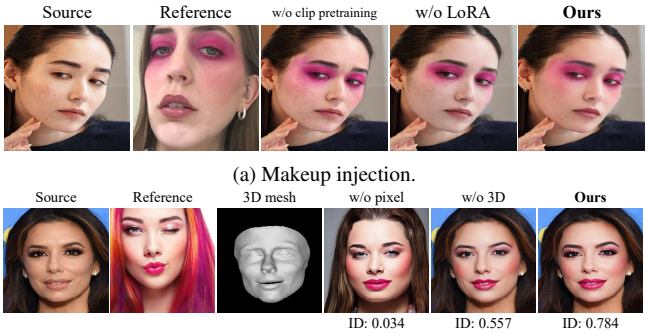



Figure 6. **Qualitative results for cross-domain faces**. Note that the reference in the last row is generated by GPT-Image-1.




(a) Makeup injection.

(b) Identity injection. The ID score between the source image and generated image is reported below the generation.

Figure 7. **Qualitative ablation on makeup/identity injection**.

line with the observations above. Fig. 7a shows that without either of the learned CLIP makeup encoder or LoRA fine-tuning, the model struggles to capture the makeup style (*i.e.*, exposed bare skin on the cheek). Fig. 7b shows that pixel-level information helps preserve the appearance while 3D mesh helps with the facial structure. Altogether, they result in the best identity preservation.

Table 4. **Ablation on identity and makeup injection modules**. **LoRA** is the LoRA fine-tuning of denoising model cross-attention layers for fusing the makeup features. **Pixel/3D** is the pixel/3D mesh guidance for the identity encoder, respectively.

| LoRA | Pixel | 3D | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ |
|---|---|---|---|---|---|---|---|
| ✓ | ✓ | × | 0.510 | 0.403 | 0.775 | 0.003 | 4.88 |
| ✓ | × | ✓ | **0.587** | 0.048 | 0.333 | 0.003 | **5.57** |
| × | ✓ | ✓ | 0.467 | **0.652** | **0.810** | 0.003 | 4.55 |
| ✓ | ✓ | ✓ | 0.528 | 0.429 | 0.797 | **0.003** | 4.95 |

Table 5. **Ablation on attention loss**. **Diff** is the diffusion loss. **Attn** is the attention loss.

| Diff | Attn | CSD ↑ | ID ↑ | SSIM ↑ | L2-M ↓ | Aes ↑ |
|---|---|---|---|---|---|---|
| ✓ | × | 0.518 | 0.416 | **0.799** | 0.003 | 4.90 |
| ✓ | ✓ | **0.528** | **0.429** | 0.797 | **0.003** | **4.95** |

#### 4.5.3. Attention loss

In Tab. 5, we study the effect of attention loss. The results show that the attention loss helps improve makeup consistency. More importantly, the visualization in Fig. 5 shows that the attention loss encourages the model to look at facial regions (*i.e.*, skin, eyes and mouth) and thus enables region-specific makeup transfer.

## 5. Conclusion

In this work, we propose **F**acial **R**egion-**A**ware **M**akeup features (**FRAM**) for diffusion-based makeup transfer. It has two stages: (1) fine-tuning CLIP to learn a makeup encoder on synthesized makeup style data. (2) construct before-and-after makeup image pairs and use them to learn to inject identity and facial region-aware makeup features for makeup transfer. Our FRAM surpasses prior methods on global and region-specific makeup transfer tasks.

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